

3(a)	C =neutrophil ; A polymorphonuclear leucocyte A basophil A eosinophil D = lymphocyte ; B / T <i>is neutral</i> E = monocyte ; A macrophage / <i>leucocyte / white blood cell, throughout</i>	3
3(b)(i)	<i>any two from:</i> lymphocytes / blood cells, produced in / originate from, bone marrow ; <i>ref. to stem cell(s) ;</i> <i>ref. to cell division ; e.g. bone marrow has cells that can carry out, mitosis / cell division</i> A (stem cells are) multipotent	2

Question	Answer	Marks
3(b)(ii)	<i>any three from:</i> <i>measles</i> <u>immune response</u> stimulated ; detail ; e.g. either <u>antigen</u> recognition / presence / binding or lymphocyte / white blood cell, stimulated / triggered plus clonal, expansion / proliferation / cell division / mitosis } or clonal selection / AW R incorrect pathogen name R bacteria <i>CLL</i> result of, uncontrolled mitosis ; A uncontrolled, division / cell cycle I rapid mitosis detail ; owing to, mutation / named mutation or cell cycle checkpoints not working or cells don't respond to signals to control division	3
3(c)(i)	<i>any two from:</i> (passive / simple) diffusion (of oxygen) across the phospholipid bilayer ; A movement from a higher to a lower concentration <i>for diffusion</i> A between phospholipids / between fatty acid chains / across hydrophobic core / across bilayer / AW, <i>for phospholipid bilayer</i> small size ; non-polar / uncharged ;	2

Question	Answer	Marks
3(c)(ii)	<i>any three from:</i> less oxygen in inhaled air / AW ; A less oxygen taken up from alveoli low(er) saturation of haemoglobin (with oxygen) ; A Hb for haemoglobin A haemoglobin carries less oxygen <i>must be the context of oxygen</i> R saturation, with air / carbon dioxide haemoglobin has lower affinity for oxygen ; (so) <u>more</u> haemoglobin required (so more red blood cells) / (more red blood cells) so <u>more</u> haemoglobin ; AW compensation / compensates ; A so same, volume / quantity / amount, of oxygen reaches body cells (compared to lower altitude)	3

Question	Answer	Marks
3(d)	<i>any four from:</i> 1 (alleles have) different nucleotide sequences ; A different base sequences 2 base substitution / substitution of a base ; A T replaced by A 3 (alleles have) different <u>mRNA</u> codons ; A altered mRNA codon R altered genetic code 4 (results in) one amino acid change or change from, glutamic acid / glutamate / glu, to, valine / val or changed primary structure ; A changed sequence of amino acids R <i>if implies whole sequence changes</i> 5 AVP ; e.g. Hb^A = CTC and Hb^S = CAC (on template strand) A Hb^A = GAG and Hb^S = GTG (on non-template strand) or Hb^A = CTT and Hb^S = CAT (on template strand) A Hb^A = GAA and Hb^S = GTA (on non-template strand) <i>this is not also mp2</i> <i>ref. to amino acid substitution at position 6 (this is not also mp4)</i> A 6th, codon / triplet, changed 6 changed, (β-globin / Hb) tertiary structure ; A polypeptide / (Hb molecule) quaternary structure R β-globin / polypeptide 7 <i>ref. to loss of globular structure / hydrophobic amino acid faces outside (instead of towards centre) ;</i> A described e.g. (val) amino acid with hydrophobic R-group instead of (glu) amino acid with hydrophilic R-group <i>accept hydrophilic / polar, amino acid</i> <i>accept hydrophobic / non polar, amino acid</i>	4

Question	Answer	Marks
3(d)	8 sticky (haemoglobin) molecule formed / forms fibres with other haemoglobin molecules ; 9 oxygen taken up less easily / less oxygen carried by haemoglobin / AW ; A β-globin takes up oxygen less easily / AW 10 AVP ; (sickle cell) haemoglobin / β-globin, less soluble / AW I insoluble	